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IMAGE HEATING DEVICE AND IMAGE FORMING APPARATUS

Abstract

An image heating device includes a heater having a plurality of heating elements which includes a first heating element group and a second heating element group, a control portion which supplies power to the first heating element group through a first common circuit and supplies power to the second heating element group through a second common circuit. The first common circuit includes a first safety element which responds to the temperature of the heating elements included in the first heating element group, and the second common circuit includes a second safety element which responds to the temperature of the heating elements included in the second heating element group. The first safety element is arranged on one side with respect to a conveyance reference position of a recording material, and the second safety element is arranged on the other side with respect to the conveyance reference position.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application is a Continuation of U.S. patent application Ser. No. 18/170,132, filed Feb. 16, 2023, which is hereby incorporated herein by reference in its entirety.

BACKGROUND OF THE INVENTION

Field of the Invention

[0002] The present invention relates to an image forming apparatus such as a printer, a copier, and the like that uses an electrophotographic method. The present invention also relates to a fixing device mounted on an image forming apparatus, or an image heating device such as a gloss imparting device that improves a gloss value of a toner image by reheating the toner image fixed to a recording material.

Description of the Related Art

[0003] A fixing device of a film heating type is known as a fixing device used in an electrophotographic image forming apparatus. In the fixing device of a film heating type, the temperature rise in a non-paper-passing portion described below is known to be a problem. The temperature rise in the non-paper-passing area is a phenomenon in which where small-size paper is continuously printed in an image forming apparatus using such a fixing device, the temperature of the area where the paper does not pass in the longitudinal direction of a nip portion gradually rises. Where the temperature of the non-paper-passing portion becomes too high, each part in the apparatus, such as a heater, a fixing film, and a pressure roller, will be damaged. Also, when large-size paper is printed while the temperature of the non-paper-passing area is rising, a phenomenon such as high-temperature offset of toner may occur in the area corresponding to the non-paper-passing portion of small-size paper.

[0004] As one means for suppressing the temperature rise in the non-paper-passing portion, a fixing device having the configuration shown in Japanese Patent Application Publication No. 2017-54071 has been proposed. That is, the fixing device includes a heater (hereinafter referred to as a split heater) in which a heating resistor (heating element) is arranged on a substrate while being divided in the longitudinal direction of the heater. By using this configuration, the heating element on the heater can be split into a plurality of heating areas (hereinafter referred to as heating blocks HB) in the longitudinal direction of the heater, and the heat generation distribution of the heater can be switched according to the size of the recording material. By doing so, it is possible to suppress the temperature rise in the non-paper-passing portion even when small-size paper is passed.

[0005] Furthermore, Japanese Patent Application Publication No. 2017-54071 proposes a configuration in which a circuit for supplying power to a plurality of heating elements is shared. That is, a configuration in which a common drive is used to supply power to a plurality of heating blocks installed left-right symmetrically with respect to the center of the paper. By adopting this configuration, it is possible to reduce the size and cost of the device and save energy.

SUMMARY OF THE INVENTION

[0006] When using a fixing device using a split heater, it is desirable to provide a safety element for each drive circuit in consideration of failure of the device. That is, it is desirable to install a

safety element in at least one of a plurality of heating blocks to which power is supplied by a common drive circuit. Where one of the drive circuits becomes uncontrollable and generates abnormal heat, a safety element installed in each drive circuit will detect the high temperature and the power supply can be quickly stopped. Here, as a safety element, a small thermostat (hereinafter referred to as a thermoswitch) is widely used from the viewpoint of function and cost.

[0007] In particular, thermoswitches that are to be arranged in narrow spaces such as the internal space of a fixing film often have to be shaped to be elongated in the longitudinal direction of the fixing film in order to function as a safety element. Since such a thermoswitch has connector terminals at both longitudinal ends of the thermoswitch, a certain margin is required as an installation space in the longitudinal direction of the fixing film. Meanwhile, depending on the number of divisions of the heating blocks of the heater, the longitudinal width of each heating block may become narrower. Therefore, when a thermoswitch is to be arranged in each of the adjacent heating blocks, the space in the longitudinal direction of the thermoswitches becomes cramped, making it difficult to arrange the thermoswitches.

[0008] An object of the present invention is to provide a technique that can ensure safety without being restricted by the device layout.

[0009] In order to achieve the above object, the image heating device of the present invention has the following: [0010] a heater having a plurality of heating elements arranged in a width direction of a recording material perpendicular to a conveyance direction of the recording material; [0011] a nip forming portion that forms a nip where the recording material is nipped; and [0012] a control portion that controls power supplied to the plurality of heating elements; [0013] wherein the image heating device heats an image formed on the recording material, which is nipped in the nip, by heat of the heater; [0014] wherein the plurality of heating elements include a first heating element group and a second heating element group, the first heating element group being arranged symmetrically with respect to a conveyance reference position of the recording material in the width direction and the second heating element group being arranged symmetrically with respect to the conveyance reference position so as to be adjacent to the first heating element group in the width direction; [0015] wherein the control portion supplies power to the first heating element group through a first common circuit, and supplies power to the second heating element group through a second common circuit; [0016] wherein in the image heating device, [0017] the first common circuit includes a first safety element arranged at the heater to respond to the temperature of the heating elements included in the first heating element group; and [0018] the second common circuit includes a second safety element arranged at the heater to respond to the temperature of the heating elements included in the second heating element group; [0019] wherein the first safety element is arranged on one side with respect to the conveyance reference position in the width direction, and [0020] wherein the second safety element is arranged on the other side with respect to the conveyance reference position in the width direction.

In order to achieve the above object, the image forming apparatus of the present invention has the following: [0021] an image forming portion that forms an image on a recording material; and [0022] a fixing portion that fixes the image formed on the recording material to the recording material; [0023] wherein the fixing portion is the image heating device of the present invention.

[0024] According to the present invention, it is possible to ensure safety without being restricted by the device layout.

[0025] Further features of the present invention will become apparent from the following description of exemplary embodiments with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0026] FIG. **1** is a schematic cross-sectional view of an image forming apparatus according to Embodiment **1**;
[0027] FIG. **2** is a cross-sectional side view of a fixing device according to Embodiment **1**;
[0028] FIGS. **3A** to **3C** are schematic cross-sectional views of a heater according to Embodiment **1**;
[0029] FIG. **4** is an electric circuit diagram according to Embodiment **1**;
[0030] FIGS. **5A** to **5C** are schematic cross-sectional views of a thermoswitch according to Embodiment **1**;
[0031] FIG. **6** is a schematic plan view of a heater and a thermoswitch according to Embodiment **1**
[0032] FIGS. **7A** and **7B** are schematic plan views of a heater and a thermoswitch according to a comparative example;
[0033] FIGS. **8A** and **8B** are schematic plan views of a heater according to a modification example of Embodiment **1**;
[0034] FIGS. **9A** and **9B** are schematic plan views of a heater and a thermal fuse according to Embodiment **2**;
[0035] FIGS. **10A** and **10B** are schematic plan views of the thermal fuse **60** according to Embodiment **2**; and
[0036] FIG. **11** is a schematic plan view of a heater according to Embodiment **1**.

DESCRIPTION OF THE EMBODIMENTS

[0037] Referring to the drawings, forms for carrying out the present invention will be described in detail by way of example on the basis of an embodiment example. Note that dimensions, materials, shapes, relative positioning, and the like of components described in each of embodiments are to be appropriately changed in accordance with configurations of an apparatus and a device to which the invention is applied and various conditions, and are therefore not intended to limit the scope of the invention to the following embodiments.

Embodiment **1**

[0038] FIG. **1** is a schematic cross-sectional view of an image forming apparatus **100** according to an embodiment of the present invention using electrophotographic recording technology. Examples of image forming apparatuses to which the present invention can be applied include copiers, printers, and the like using an electrophotographic method or an electrostatic recording method. A case of application to a laser printer that forms an image on a recording material **P** such as recording paper by using an electrostatic recording method will be described herein.

Schematic Configuration of Image Forming Apparatus

[0039] FIG. **1** is a schematic cross-sectional view of an example of an image forming apparatus according to Embodiment **1** of the present invention. This image forming apparatus includes an image forming portion **A** that forms a toner image on a recording material, a recording material feeding portion **B** that feeds the recording material to the image forming portion **A**, and a fixing portion (fixing device) **C** that heats and fixes a toner image on the recording material to the recording material. The image forming portion **A** has a drum-type electrophotographic photosensitive member (hereinafter referred to as a photosensitive drum) **1** as an image bearing member. The photosensitive drum **1** is rotatably supported by an apparatus main body **M** of the image forming apparatus **100** constituting a housing of the image forming apparatus. A charging roller **2**, a laser scanner **3**, a developing device **4**, a transfer roller **5**, and a cleaning device **6** are arranged in this order around the outer peripheral surface of the photosensitive drum **1** along the rotation direction thereof. The recording material feeding portion **B** has a delivery roller **11**. The delivery roller **11** is rotated in the direction of the arrow at a predetermined timing by a conveyance drive motor (not shown) to deliver the recording material **P** stacked and accommodated in a cassette **7** to a conveyance path.

[0040] The image forming apparatus of Embodiment **1** has a control portion (not shown) that

controls the image forming portion A, the recording material feeding portion B, the fixing device C, and the like. The control portion is composed of a CPU and a memory such as ROM and RAM, and various programs required for image formation are stored in the memory. This control portion receives a print signal from an external device such as a host computer and executes a predetermined image formation control sequence on the basis of the print signal. As a result, the drum motor is driven to rotate, and the photosensitive drum **1** rotates in the direction of the arrow at a predetermined peripheral speed (process speed). The surface of the rotated photosensitive drum **1** is uniformly charged to a predetermined potential having the same polarity as the toner (negative polarity in this case) by the charging roller **2**. The laser scanner **3** scans the charged surface of the photosensitive drum **1** with a laser beam L on the basis of image information to expose the surface of the photosensitive drum **1**. As a result of the exposure, the charge is removed from the exposed portion and an electrostatic latent image is formed on the surface of the photosensitive drum **1**.

[0041] The developing device **4** has a developing roller **41** and a toner container **42** accommodating the toner. The toner is rubbed by a member such as a urethane blade (not shown) and charged to a predetermined polarity (negative polarity in Embodiment 1). When a negative voltage is applied to the developing roller **41** from a developing voltage power source (not shown) in the developing device **4**, the toner is caused to adhere to the electrostatic latent image on the surface of the photosensitive drum **1** by using a potential difference, and the latent image is developed as a toner image T. The toner image T formed on the surface of the photosensitive drum **1** is transferred to the recording material P by applying a positive voltage having a polarity opposite to that of the toner to the transfer roller **5** and utilizing the potential difference created by the transfer voltage. Further, the conveyance drive motor provided in the recording material feeding portion B is rotationally driven, and the delivery roller **11** delivers the recording material P from the cassette **7** to the conveyance roller **8**. The recording material P is conveyed by the conveyance roller **8**, passes a top sensor **9**, and is conveyed to a transfer nip portion between the surface of the photosensitive drum **1** and the outer peripheral surface of the transfer roller **5**. The recording material P onto which the toner image formed on the surface of the photosensitive drum **1** has been transferred is conveyed along a conveyance guide **10** to the fixing device C, where the toner image on the recording material P is heated and pressed to be thermally fixed on the recording material P. The recording material P on which the toner image T has been thermally fixed is conveyed by the conveyance roller **12** and a discharge roller **13** in this order and is discharged to a discharge tray **14** on the upper surface of the apparatus main body M. Untransferred toner remaining on the surface of the photosensitive drum **1** after the toner image has been transferred onto the recording material P is removed by a cleaning blade **64** of the cleaning device **6** and accumulated in the cleaning device **6**. By repeating the above operations, sequential printing is performed. The image forming apparatus of Embodiment 1 can print at a printing speed of 70 sheets/min for A4 size. Although the details are omitted, the image forming apparatus of Embodiment 1 is provided with a reversing conveyance path that enables double-sided image formation, and the apparatus is configured so that the recording material P having an image formed on one side is returned to the upstream side of the image forming portion A by switchback caused by reverse rotation of the discharge roller **13**.

Configuration of Fixing Device

[0042] FIG. **2** is a schematic cross-sectional side view of the fixing device C as an image heating device according to Embodiment 1. The fixing device C of Embodiment 1 basically includes a heater **1100**, a heater holder **29**, a metal stay **22**, a fixing film **25** as a tubular film that is a fixing member, and a pressure roller **26**. The heater holder **29** is a holding member that holds (supports) the heater **1100** as a heating body inside the fixing film **25**. In the fixing device C, the recording material P is nipped and the toner image T is thermally fixed to the recording material P by using the heat of the heater **1100** in a nip portion N between the tubular fixing film **25** as a heating rotating body and the pressure roller **26** as a pressing rotating body (pressure member). The nip portion N is formed by the heater **1100** and the pressure roller **26** with the fixing film **25** interposed

therebetween. The recording material P is nipped and conveyed at the nip portion N by the rotation of the pressure roller **26** and the driven rotation of the fixing film **25**. In the present embodiment, the heater **1100** is configured to be in direct contact with the inner surface of the fixing film **25**, but a heat transfer member or the like may be interposed between the heater **1100** and the inner surface of the fixing film **25**. Of the constituent members of the fixing device C according to the present embodiment, the members related to the formation of the nip portion N constitute a nip forming portion. A power application control portion **421** connected to a commercial AC power supply supplies power to the fixing device C according to a signal from the CPU **420**.

Pressure Roller

[0043] The pressure roller **26** has an elastic layer **262** on the outer circumference of a core shaft portion **261** and has a surface layer **263** on the outer circumference of the elastic layer **262**. The pressure roller **26** has an outer diameter of about 25 mm. A solid or hollow metal material such as aluminum or iron is used for the core shaft portion **261**. In Embodiment 1, solid aluminum is used as the core metal material. The elastic layer **262** is composed of heat-resistant silicone rubber and made conductive by adding an electrically conductive material such as carbon. The surface layer **263** in contact with the outer surface of the fixing film **25** is a releasing tube having a thickness of 10 μm to 80 μm and made of a fluorine resin such as PFA, PTFE, FEP, and the like. Here, PFA stands for a tetrafluoroethylene/perfluoroalkyl vinyl ether copolymer, PTFE for polytetrafluoroethylene (tetrafluoro), and FEP for fluoroethylene/hexafluoropropylene copolymer (tetrafluoro/hexafluoro). The surface layer **263** is desirably made conductive from the viewpoint of preventing charge-up associated with the passage of paper. In Embodiment 1, the surface layer **263** of the pressure roller **26** was configured by adding carbon as a conductive material to a PFA tube having a thickness of 30 μm .

Fixing Film

[0044] The fixing film **25** has a cylindrical shape with a diameter of 24 mm. The fixing film **25** is flexible and loosely fitted on the heater holder **29**. The layer structure of the fixing film **25** is composed of multiple layers including a base layer **251**, an elastic layer **252**, and a surface layer **253** from the inside, as in the cross-sectional configuration shown in the circle in FIG. 2. As a material for the base layer **251**, a heat-resistant resin material with low heat capacity, such as a polyimide, a polyamideimide, PEEK, or PES, is generally used. Metal materials such as SUS may also be used. Since the base layer **251** needs to have a small heat capacity to satisfy the quick start property and at the same time to satisfy the mechanical strength, the thickness of 18 μm or more and 150 μm or less is desirable. The base layer **251** of Embodiment 1 was a cylindrical polyimide base layer with a thickness of 70 μm . The elastic layer **252** is made of an elastic material typified by silicone rubber. By providing the elastic layer **252**, it is possible to wrap the toner image T and apply heat uniformly, so that it is possible to obtain a high-quality image without unevenness. Since the elastic layer **252** has low thermal conductivity with silicone rubber alone, a thermally conductive filler such as alumina, metallic silicon, silicon carbide, zinc oxide, or the like is added to impart high thermal conductivity. In a high-speed machine as in Embodiment 1, the amount of the thermally conductive filler to be added may be adjusted, as appropriate, to ensure 0.9 W/m.K or more. In Embodiment 1, alumina and metallic silicon are added as thermally conductive fillers to the rubber material of the elastic layer **252**, and the thermal conductivity thereof is set to 1.5 W/m.K. Also, the thickness of the elastic layer **252** is set to 270 μm . As a release layer, the surface layer **253** is required to have high toner releasability and high abrasion resistance. A fluororesin such as PFA, PTFE, and FEP is used as the material. The layer is formed of a coating layer obtained by baking a resin dispersion, or a tube layer. An additive such as carbon and an ion-conducting material may be added to the fluororesin to impart electrical conductivity. The surface layer **253** of Embodiment 1 was a tube layer having a thickness of 20 μm and using PFA as a fluororesin, with no conductive material being added thereto.

Heater Holder

[0045] The heater **1100** is held by the heater holder **29** made of a heat-resistant resin material such as a liquid crystal polymer. The heater holder **29** also has a guide function of guiding the rotation of the fixing film **25**.

Heater

[0046] The heater **1100**, which is a characteristic configuration of Embodiment 1, will be described with reference to FIGS. **3A** to **3C**. The heater **1100** has a ceramic substrate **1105** and a heating resistor (heating element) provided on the substrate **1105** and generating heat by application of electric power. A surface protective layer **1108** made of glass is provided on the surface (first surface) of the substrate **1105** that contacts the fixing film **25** on the nip portion N side in order to ensure slidability with the fixing film **25**. A surface protective layer **1107** made of glass is provided on the surface (second surface) of the substrate **1105** opposite to the first surface on the nip portion N side to insulate the heating resistor. An electrode **E14** is exposed on the second surface, and an electric contact **C14** for power supply is in contact with the electrode to connect the heating resistor electrically to the AC power supply.

[0047] FIGS. **3A** and **3B** show the configuration of the heater **1100** of the present embodiment. FIG. **3A** is a cross-sectional view of the heater **1100** at a conveyance reference position X of the recording material P shown in FIG. **3B**. FIG. **3B** is a schematic plan view of each layer of the heater **1100**. FIG. **3C** is a schematic plan view of the heater holder that holds the heater **1100**. In the present embodiment, the conveyance reference position X is set at a substantially central position in the width direction of the recording material perpendicular to the conveyance direction of the recording material P in the fixing device C, but the set position is not limited to a specific position.

[0048] The heater **1100** is configured of the substrate **1105**, a sliding surface layer **1** provided on the first surface side of the substrate **1105** that contacts the fixing film **25**, a sliding surface layer **2** that covers the sliding surface layer **1**, a back surface layer **1** provided on the second surface side of the substrate **1105** opposite to the first surface side, and a back surface layer **2** that covers the back surface layer **1**. The heater **1100** has a plurality of heating blocks configured of a first conductor **1101**, a second conductor **1103**, and a heating element **1102** on the back surface layer **1** along the longitudinal direction. In the heater **1100** of the present embodiment, a total of five heating blocks HB**11** to HB**15** are formed by a plurality of heating elements **1102** arranged in the width direction (longitudinal direction of the substrate **1105**) perpendicular to the recording material conveyance direction.

[0049] The heater **1100** shown in FIGS. **3A** to **3C** are divided into heating blocks HB**11** to HB**15** left-right symmetrically (symmetrically with respect to the conveyance reference position X) from the center of the heater **1100** in the longitudinal direction of the heater (width direction of the recording material perpendicular to the recording material conveyance direction). The division positions of the heating block HB correspond to “A5 size”, “B5 size” and “A4 size”, respectively. That is, the width of the heating block HB**13** is 150 mm, which is substantially the same as the short side length of A5 size. The width of the heating blocks HB**12** to HB**14** is 182 mm, which is substantially the same as the short side length of B5 size. The width of the heating blocks HB**11** to HB**15** is 210 mm, which is substantially the same as the short side length of A4 size. Among these heating blocks HB, “the heating block HB**13**” is designated as a heating group 1, “the heating block HB**12** and the heating block HB**14**” are designated as a heating group 2, and “the heating block HB**11** and the heating block HB**15**” are designated as a heating group 3. Each heating group is powered by the same drive (common circuit).

[0050] The heating elements **1102** of each heating block are arranged while being divided into a heating element **1102a** on the upstream side and a heating element **1102b** on the downstream side in the recording material P passage direction in relation to the widthwise direction (direction perpendicular to the longitudinal direction) of the heater **1100**. Also, the first conductor **1101** is divided into a conductor **1101a** connected to the heating element **1102a** and a conductor **1101b** connected to the heating element **1102b**.

[0051] The heater **1100** is divided into five heating blocks **HB11** to **HB15**. That is, the heating element **1102a** is divided into five heating elements **1102a-1** to **1102a-5**. Similarly, the heating element **1102b** is divided into five heating elements **1102b-1** to **1102b-5**. Further, the second conductor **1103** is also divided into five conductors **1103-1** to **1103-5**.

[0052] The back surface layer **2** of the heater **1100** is provided with an insulating surface protective layer **1107** that covers the heating elements **1102**, the first conductors **1101** and the second conductors **1103**. In Embodiment 1, glass is used as the surface protective layer **1107**. The surface protective layer **1107** does not cover the electrodes **E11** to **E15**, **E18-1** and **E18-2** with which the electric contacts **C11** to **C15**, **C18-1** and **C18-2** for power supply are in contact. The electrodes **E11** to **E15** are for supplying power to the heating blocks **HB11** to **HB15** via the second conductors **1103-1** to **1103-5**, respectively. The electrodes **E18-1** and **E18-2** are for supplying electric power to the heating blocks **HB11** to **HB15** via the first conductors **1101a** and **1101b**.

[0053] Where the electrodes are thus provided on the back surface of the heater **1100**, it is not necessary to provide a conductive pattern for supplying power to the second conductors **1103-1** to **1103-5** on the substrate **1105**, so that the width of the substrate **1105** in the widthwise direction can be reduced. As a result, an increase in the size of the heater **1100** can be suppressed. As shown in FIG. 3B, the electrodes **E12** to **E14** are arranged in the longitudinal direction of the substrate **1105** in the area where the heating elements are provided.

[0054] Although details will be described hereinbelow, various heat generation distributions can be formed in the heater **1100** of Embodiment 1 by independently controlling a plurality of heating blocks. A heat generation distribution corresponding to the size of the recording material **P** can thus be set. Furthermore, the heating elements **1102** are formed of a material having PTC (Positive Temperature Coefficient) properties. By doing so, even in the case where the end portion of the recording material **P** does not coincide with the boundary of the heating block, it is possible to suppress the temperature rise in the non-paper-passing portion.

[0055] A plurality of thermistors (temperature detection elements) for detecting the temperature of each of the heating blocks **HB11** to **HB15** is formed on the sliding surface layer **1** on the sliding surface side (the surface that contacts the fixing film **25**) of the heater **1100**. The plurality of thermistors are indicated by **T11-1C** to **T11-3C**, **T11-1E** to **T11-3E**, **T12-4C** to **T12-5C**, and **T12-3E** to **T12-7E**, respectively, in FIG. 3B. The thermistor material may be any material that has a high positive or negative value of TCR (Temperature Coefficient of Resistance). In Embodiment 1, the thermistor was configured by thinly printing a material having NTC (Negative Temperature Coefficient) on the substrate **1105**.

[0056] The thermistor arrangement for each heating block **HB** will be explained hereinbelow. In Embodiment 1, as shown in FIG. 3B, two or more thermistors are arranged in all of the heating blocks **HB11** to **HB15**. For example, two thermistors **T11-2C** and **T11-2E** are installed for the heating block **HB12** to obtain a configuration in which the respective temperature can be detected by conductive patterns **ET11-2C** and **ET11-2E** for resistance value detection and a common conductive pattern **EG11**. The thermistor **T11-2C** is a main thermistor for detecting the temperature of the central area of the heating block **HB12**, and is arranged substantially at the center of the heating block **HB12** in the width direction of the recording material **P**. The thermistor **T11-2E** is an end portion thermistor for detecting the temperature of the end portion area of the heating block **HB12**, and is arranged in the area of the heating block **HB12** and on the side adjacent to the heating block **HB11** in the width direction of the recording material **P**. In this way, main thermistors **T11-1C**, **T11-2C**, **T11-3C**, **T12-4C**, and **T12-5C** for detecting the temperature of the central area are arranged for each of the heating blocks **HB11** to **HB15**. Further, end portion thermistors **T11-1E**, **T11-2E**, **T11-3E**, **T12-3E**, **T12-4E**, and **T12-5E** for detecting the temperature of the end portion areas are arranged for each of the heating blocks **HB11** to **HB15**.

[0057] The heater holder **29** will be described with reference to FIG. 3C. The heater holder of the present embodiment is provided with opening portions **HC11** to **HC15**, **HC18-1** and **HC18-2** for

supplying power to the electrodes E11 to E15, E18-1 and E18-2 provided on the back surface layer 1 of the heater 1100. Electrical contacts C11 to C15, C18-1 and C18-2 as power supply portions supply power to the electrodes through the opening portions HC11 to HC15, HC18-1 and HC18-2. The heater holder 29 is also provided with opening portions H212-12, H212-13, and H212-15 for installing thermostats 50, which will be described hereinbelow.

Power Application Control Circuit for Heater

[0058] FIG. 4 is a circuit diagram of a control circuit 1400 that controls the heater 1100. Power control (power application control) for the heater 1100 is performed by conducting/interrupting power supply to the heater 1100 by triacs 1411 to 1413. The triacs 1411 to 1413 operate according to FUSER11 to FUSER13 signals from a CPU 420, respectively. The control circuit 1400 of the heater 1100 has a circuit configuration capable of applying power to the five heating blocks HB11 to HB15 by three triacs 1411 to 1413. Specifically, the triac 1411 performs power application control of the heating block HB11 and the heating block HB15 (heating group 3). In addition, the triac 1412 performs power application control of the heating block HB12 and the heating block HB14 (heating group 2). In addition, the triac 1413 performs power application control of the heating block HB13 (heating group 1) as a reference heating element group. At this time, the heating group 2 as the first heating element group and the heating group 3 as the second heating element group each have a plurality of heating blocks HB in one drive circuit. As shown in FIG. 4, power is supplied to each of the heating elements included in the group via one respective common drive circuit (first common circuit, second common circuit). With respect to the reference points A-F in FIG. 4, Segment A-B is a common circuit for all heating elements, Segment B-C is a common circuit for the 1st heating element group, Segment D-E is a common circuit for the 2nd heating element group, and Segment D-F is a circuit for the central heating element. In FIG. 4, drive circuits for the triacs 1411 to 1413 are omitted.

[0059] A zero-cross detection portion 1421 is a circuit that detects a zero-cross of an AC power supply 1401 and outputs a ZeroX signal to the CPU 420. The ZeroX signal is used as a reference signal or the like for phase-controlling the triacs 1411 to 1413.

Heater Temperature Detection Method

[0060] A method for detecting the temperature of the heater 1100 will be described hereinbelow. Signals (Th11-1C to Th11-3C, Th11-1E to Th11-3E, Th12-4C to Th12-5C, and Th12-3E to Th12-5E) obtained by dividing a voltage Vcc by the resistance value of the thermistors and the resistance values of resistors 1451 to 1461 are input to the CPU 420. In FIG. 4, the thermistors are denoted by T11-1C to T11-3C, T11-1E to T11-3E, T12-4C to T12-5C, and T12-3E to T12-5E. For example, the signal Th11-3C is obtained by dividing the voltage Vcc by the resistance value of the thermistor T11-3C and the resistance value of the resistor 1456. Since the thermistor T11-3C has a resistance value corresponding to the temperature, the level of the signal Th11-3C input to the CPU 420 also changes when the temperature of the heating block HB13 changes. The CPU 420 converts each input signal into a temperature corresponding to the level thereof.

[0061] The CPU 420 calculates the electric power to be supplied to the heater, for example, by PI control on the basis of the set temperature (control target temperature) of each heating block and the detected temperature (output) of each thermistor. Furthermore, the calculated supply power is converted into control timings such as corresponding phase angles (phase control) and fractions (fraction control), and the triacs 1411 to 1413 are controlled at these control timings. Processing of signals corresponding to other thermistors is the same, so the description thereof is omitted.

Heater Power Control

[0062] The power control of the heater 1100 (heater temperature control) will be described hereinbelow. During the fixing process, each of the heating blocks HB11 to HB15 is controlled so that the temperature detected by the thermistors is maintained at the set temperature (control target temperature). Specifically, the power supplied to the heating block HB14 is determined by controlling the driving of the triac 1414 so that the temperature detected by the thermistor T11-4C

is maintained at the set temperature. Thus, each thermistor is used when performing control for keeping each heating block HB at a constant temperature.

Heater Protection

[0063] A relay **1440** is provided as a means for cutting off power to the heater **1100** when the temperature of the heater **1100** rises excessively due to factors such as device failure. Also, three thermostiches **50-12**, **50-13**, and **50-15** are on the DC circuit connected to the **24 V** power supply. In this configuration, where any one of the three thermostiches **50-12**, **50-13**, and **50-15** is opened, the **24 V** applied to the relay **1440** is cut off, the relay **1440** is opened, and the AC circuit is cut off.

Thermoswitch

[0064] The safety element, which is a characteristic configuration of Embodiment **1**, will be explained hereinbelow. The safety element is installed for the purpose of safely shutting off the power input to the heater **1100** without damaging the heater **1100** or the like even when the CPU **420** cannot perform control and the heater **1100** runs out of control. In the present embodiment, a case where a thermostich is used as a safety element will be described, but other elements such as a thermostat and a thermal fuse may be used as the element that detects abnormal heat generation of the heater and cuts off the power supply to the heater.

[0065] The operation of the thermostich **50** (**50-12**, **50-13**, and **50-15**) will be described with reference to FIGS. **5A** to **5C**. FIGS. **5A** to **5C** illustrate schematic configurations of the thermostich **50** used as a safety element in Embodiment **1**. FIG. **5A** is a schematic diagram showing a cross section of the thermostich **50** perpendicular to the plane of the heater **1100** and along the longitudinal direction of the heater **1100** in a state where the thermostich **50** is not in operation, that is, the circuit is connected. FIG. **5B** is a schematic cross-sectional view (similar to FIG. **5A**) of the thermostich **50** in a state where the predetermined temperature has been exceeded and the thermostich **50** operated, that is, in a state where the circuit has been disconnected. FIG. **5C** is a plan view of the thermostich **50**.

[0066] As shown in FIG. **5A**, the thermostich **50** is composed of a heat-sensitive portion **51** that senses the temperature of the heater **1100** or the like, a ceramic body **52**, and a connector terminal **53**. A bimetal **55**, a metal rod **56**, a movable contact **57** and a contact portion **58** are arranged inside the heat-sensitive portion **51**. The bimetal **55** is formed by bonding two metal plates with different coefficients of thermal expansion into a curved shape and has a property of deforming into a curved shape in the opposite direction when the temperature of the bimetal **55** exceeds a predetermined value. The thermostich **50** is connected through a connector terminal **53** to the drive circuit of the heating group in which the thermostich is installed. Connector terminals are provided at both longitudinal ends of the ceramic body **52** (both ends in the longitudinal direction of the heater).

[0067] As shown in FIG. **5B**, when the temperature of the location in the heater **1100** where the thermostich **50** is installed exceeds a predetermined temperature, the bimetal **55** deforms into a shape curved in the opposite direction. This deformation of the bimetal **55** pushes down the metal rod **56** and further pushes down the movable contact **57**. As a result, the contact portion **58** separates and the circuit is disconnected. The thermostich **50** used in Embodiment **1** has a size of $Xa \times Ya$: $36\text{ mm} \times 8\text{ mm}$, which is widely used.

Comparative Example

[0068] A heater according to a comparative example will be described with reference to FIGS. **7A** and **7B**.

[0069] FIG. **7A** is a plan view of a heater according to Comparative Example **1**, explaining the configuration of the heater and the installation position of thermostiches according to Comparative Example **1**. The heater according to Comparative Example **1** is a heater to be used in a fixing device with a maximum print width of A3 size, and this heater is configured to have a longitudinal width wider than that of the heater according to the present embodiment, which is used in a fixing device with a maximum print width of A4 size.

[0070] In the split heater according to Comparative Example 1 shown in FIG. 7A, the heater is divided into the heating blocks HB11 to HB15 left-right symmetrically from the longitudinal center of the heater (conveyance reference position). The division positions of the heating blocks HB11 to HB15 correspond to “A5 size”, “A4 size”, and “A3 size”. That is, the width of the heating block HB13 is 150 mm, which is substantially the same as the short side length of A5 size. Further, the width of the heating blocks HB12 to HB14 is 210 mm, which is substantially the same as the short side length of A4 size. Furthermore, the width of the heating blocks HB11 to HB15 is 297 mm, which is substantially the same as the short side length of A3 size. Further, among these heating blocks HB11 to HB15, “heating block HB13=heating group 1”, “heating block HB12 and heating block HB14=heating group 2”, and “heating block HB11 and heating block HB15=heating group 3”, and each group is powered by the same drive circuit (common circuit).

[0071] Thermoswitches 510-11, 510-12, and 510-13 have the same configuration as the thermoswitch 50 according to the present embodiment shown in FIGS. 5A to 5C.

[0072] From the viewpoint of safety, it is desirable to install the thermoswitches 510-11, 510-12, and 510-13 each in one of the heating blocks HB belonging to the heating group powered by the same drive. As an example of the installation configuration, as shown in FIG. 7A, a configuration can be considered in which thermoswitches 510-11, 510-12, and 510-13 are installed in the heating block HB11, the heating block HB12, and the heating block HB13, respectively. That is, in this configuration all the thermoswitches are installed in the heating blocks HB located on one longitudinal side with respect to the conveyance reference position which is the center of the paper width.

[0073] In the heater of Comparative Example 1, which is used in a fixing device with a maximum print width of A3 size, all the thermoswitches can be installed in the heating blocks HB on one longitudinal side with respect to the conveyance reference position in order to ensure a sufficient width of the heating blocks HB. However, in a heater having a heating block HB with a small width, such as the heater according to the present embodiment, it may be difficult to install the thermoswitches due to space considerations.

[0074] FIG. 7B is a plan view of a heater according to Comparative Example 2 for explaining the configuration of the heater and the thermoswitch installation positions according to Comparative Example 2. The heater according to Comparative Example 2, like the heater according to the present embodiment, is to be used in a fixing device having a maximum print width of A4 size, and this heater is configured to have a longitudinal width of the heating blocks HB narrower than that of the heater according to Comparative Example 1.

[0075] As shown in FIG. 7B, in the heater according to Comparative Example 2, the splitting positions of the heating blocks HB correspond to “A5 size”, “B5 size”, and “A4 size” in the same way as in the heater according to the present embodiment. That is, the width of the heating block HB13 is 150 mm, which is substantially the same as the short side length of A5 size. Further, the width of the heating blocks HB12 to HB14 is 182 mm, which is substantially the same as the short side length of B5 size. Furthermore, the width of the heating blocks HB11 to HB15 is 210 mm, which is substantially the same as the short side length of A4 size. Further, among these heating blocks HB11 to HB15, “heating block HB13=heating group 1”, “heating block HB12 and heating block HB14=heating group 2”, and “heating block HB11 and heating block HB15=heating group 3”, and each group is powered by the same drive circuit (common circuit).

[0076] The thermoswitches 510-11, 510-12, and 510-13 have the same configuration as the thermoswitch 50 according to the present embodiment shown in FIGS. 5A to 5C.

[0077] The arrangement of the thermoswitches in the heater according to Comparative Example 2 is similar to that of the heater according to Comparative Example 1 in that all the thermoswitches are installed in the heating blocks HB located on one longitudinal side with respect to the conveyance reference position which is the center of the paper width. That is, in this configuration, thermoswitches 510-11, 510-12, and 510-13 are installed in the heating block HB11, the heating

block HB12, and the heating block HB13, respectively. Here, the longitudinal widths of the heating block HB11 and the heating block HB12 in the heater according to Comparative Example 2 are narrower than the longitudinal widths of the heating block HB11 and the heating block HB12 in the heater according to Comparative Example 1.

[0078] Therefore, as shown in FIG. 7B, the thermoswitch 510-11 installed in the heating block HB11 and the thermoswitch 510-12 installed in the heating block HB12 interfere with each other in the longitudinal direction of the heater at k in the figure. Since the thermoswitch has connector terminals at both longitudinal ends, a certain amount of space is required in the longitudinal direction as an installation space. Therefore, in a heater configuration in which each heating block HB has a narrow longitudinal width, as in Comparative Example 2, the installation space for adjacent heating thermoswitches becomes cramped, making it difficult to arrange the thermoswitches.

[0079] Although a comparison between the fixing device with a maximum print width of A4 size and the fixing device with a maximum print width of A3 size is shown here, similar problems occur in other configurations as well. For example, the same problem occurs in a fixing device in which the maximum print size remains A3 and the heating block HB is split into more section to support a large number of paper sizes. Installation Positions of Thermoswitches in Embodiment 1

[0080] FIG. 6 shows the installation positions of the thermoswitches 50 of Embodiment 1. In Embodiment 1, the thermoswitches 50 were installed so as not to be adjacent to each other in the left-right direction with respect to the center (conveyance reference position) X in the width direction of the recording material P in relation to the adjacent heating groups including a plurality of heating blocks HB. That is, in Embodiment 1, thermoswitches 50-12, 50-13, and 50-15 were installed in the heating blocks HB12, HB13, and HB15, respectively. By doing so, the thermoswitches 50-12, 50-13, and 50-15 can be installed without interfering with each other, and the thermoswitch 50 can be arranged for each drive circuit.

[0081] In Embodiment 1, the installation positions of the thermoswitches 50 are in the three heating blocks HB12, HB13, and HB15, but configuration examples in which the thermoswitches 50 are installed so as not to be adjacent to each other in the heating group including a plurality of heating blocks HB are not limited to this. For example, the combination of heating blocks in which the thermoswitches 50 are installed may be the heating blocks HB11, HB13, and HB14.

Modification Example

[0082] In the present embodiment, an example of the heater 1100 in which the heating element 1102a and the heating element 1102b are separately provided in the conveyance direction of the recording material P has been described, but the shape of the heating elements does not matter as long as the heater is split into heating blocks HB in the width direction of the recording material P. In addition, the configuration in which the electrodes E11 to E15, E18-1 and E18-2 are formed on the back surface of the recording material passing area of the heater 1100 is exemplified in Embodiment 1, but a configuration in which end electrodes are formed as in the modification example shown in FIG. 8A may be employed as well.

[0083] FIGS. 8A and 8B show the configuration of the heater according to the modification example. The heating area of the heater 1100-1 in FIG. 8A is divided into five heating blocks HB11 to HB15. Correspondingly, heating elements 1102-1, 1102-2, 1102-31, 1102-32, 1102-33, 1102-34, 1102-35, 1102-4, and 1102-5 are arranged. The heating blocks HB are divided into three heating groups for each drive circuit: a heating group 1 (heating block HB13), a heating group 2 (heating blocks HB12 and HB14), and a heating group 3 (heating blocks HB11 and HB15). The heating group 1 is a heating area including a reference position of the recording material P, and includes five heating elements 1102-31, 1102-32, 1102-33, 1102-34, and 1102-35 as a reference heating element group. The heating group 2 has heating blocks HB12 and HB14 installed separately on the left and right sides with the reference position of the recording material P interposed therebetween, and the heating blocks are installed so as to be adjacent to the heating group 1 on both longitudinal

sides. The heating group 2 as the first heating element group includes two heating elements **1102-2** and **1102-4**. The heating group 3 has heating blocks **HB11** and **HB15** installed separately on the left and right sides with the recording material P reference position interposed therebetween, and the heating blocks are installed so as to be adjacent to the heating group 2 on the sides away from the recording material P reference position. The heating group 3 as a second heating element group includes two heating elements **1102-1** and **1102-5**.

[0084] Further, the heating elements **1102-1** to **1102-5** have a shape folded multiple times in the width direction of the heater **1100-1**, as shown in FIG. 8A. In addition, the heating elements **1102-1** to **1102-5** generate heat by being supplied with electric power from the electrodes **E21** to **E24** via the conductors **1101a** and **1101b-1** to **1101b-5**.

[0085] Thermoswitches **50-11**, **50-13**, and **50-14** are installed at the positions shown in FIG. 8B with respect to the heater **1100-1**. That is, the thermoswitch **50-11** is installed in the area of the heating block **HB11**, the thermoswitch **50-13** is installed in the heating block **HB13**, and the thermoswitch **50-14** is installed in the heating block **HB14**. By doing so, an effect similar to that of Embodiment 1 can be obtained.

[0086] Further, in this modification example, as in Embodiment 1, the heating blocks **HB12**, **HB13**, and **HB15** can be set as a combination of heating blocks in which the thermoswitches are installed.

[0087] Also, in the present embodiment, the case in which the heating group is divided into three has been described, but the same effect can be exhibited even in a fixing device in which the heating area is split into larger number of sections. An example is shown in FIG. 11. According to FIG. 11, the heating group n is composed of a heating block **HBn** and a heating block **HBna** to which power is supplied by the same drive, and the heating block **HBn** and the heating block **HBna** are arranged separately on the left and right sides with the conveyance reference position X of the recording material P interposed therebetween. Here, n for the heating block HB indicates the number of the position at which the heating block HB is arranged on one side (left side) in the width direction of the heater with respect to the heating block HB including the conveyance reference position X. Further, na for the heating block HB indicates the number of the position at which the heating block HB is arranged on the other side (right side) in the width direction of the heater with respect to the heating block HB including the conveyance reference position X. The heating group n+1 is composed of a heating block **HB(n+1)** and a heating block **HB(n+1)a** to which power is supplied by the same drive. The heating block **HB(n+1)** and the heating block **HB(n+1)a** are arranged separately on the left and right sides with the conveyance reference position X of the recording material P interposed therebetween. Further, the heating group n+1 is installed adjacent to the heating group n in the direction away from the conveyance reference position X of the recording material P. At this time, the thermoswitches **50** are installed at the positions shown in FIG. 11. That is, the thermoswitch **50-n** is installed in the heating block **HBn**, and the thermoswitch **50-(n+1)** is installed in the heating block **HB(n+1)a**. By doing so, even in a fixing device using the heater **1100** split into multiple sections, the thermoswitches **50** can be arranged for each drive circuit without being restricted by the device layout, and problems of productivity improvement in the case of small-size paper can be solved.

[0088] Furthermore, in Embodiment 1, the heat-sensitive portion **51** of the thermoswitch **50** is arranged so as to contact the heating element **1100**, but such contact is not necessary, provided that the heat-sensitive portion can be arranged at a position making it possible to detect the temperature of the area of the heating block HB (to react to excessive temperature rise). For example, it is possible to insert a resin material or the like between the heater **1100** and the heat-sensitive portion **51**.

Embodiment 2

[0089] The fixing device according to Embodiment 2 of the present invention is configured as a fixing device operating at a low printing speed, but capable handling a finer paper size.

[0090] In the fixing device according to Embodiment 2, the maximum recording material width

that can pass is the LTR size (216 mm), and the heating block HB is split into seven heating blocks. Also, as a safety element, a thermal fuse **60** smaller than the thermoswitch **50** used in Embodiment **1** is used. The difference between Embodiment 2 and Embodiment 1 is only in the heater **1200** and the safety element (thermal fuse **60**). The rest of the configuration is the same as that of Embodiment 1, and therefore the description thereof is omitted.

[0091] The configuration of the heater **1200** of Embodiment 2 will be described with reference to FIGS. **9A** and **9B**. In the heater **1100** of Embodiment 2, the heating block HB is divided into seven heating blocks HB**21** to HB**27**. The heating block HB**24** corresponds to A5 size (150 mm). The heating block HB**23** and the heating block HB**25** correspond to the Executive size (185 mm). The heating block HB**22** and the heating block HB**26** correspond to A4 size (210 mm). The heating block HB**11** and the heating block **17** correspond to the LTR size (216 mm).

[0092] As with the heater of Embodiment 1, the heater of Embodiment 2 is also divided into heating groups for each heating block HB having a common drive circuit. A heating group 1 as a reference heating element group is composed of a heating block HB**14**. A heating group 2 as a first heating element group is composed of heating blocks HB**13** and HB**15**. A heating group 3 as a second heating element group is composed of heating blocks HB**12** and HB**16**. A heating group 4 as a third heating element group is composed of heating blocks HB**11** and HB**17**.

[0093] A thermal fuse **60**, which is a safety element used in Embodiment 2, will be described with reference to FIGS. **10A** and **10B**. FIGS. **10A** and **10B** are schematic cross-sectional views showing a schematic configuration of the thermal fuse **60**. As shown in FIG. **10A**, the thermal fuse **60** includes a resin cover **61**, a heat-sensitive fusible body **62** and a lead wires **63**. FIG. **10B** is a schematic cross-sectional view taken when the thermal fuse **60** operates. When the thermal fuse **60** is exposed to a temperature equal to or higher than a predetermined temperature, the heat-sensitive fusible body **62** is melted and the circuit is disconnected, as shown in FIG. **10B**. The thermal fuse **60** is connected via the lead wires **63** to a drive circuit of the heating group in which the thermal fuse is installed. The lead wires **63** are provided at both longitudinal ends of the resin cover **61** (both ends in the width direction of the heater). The thermal fuse **60** used in Embodiment 2 has a size of Xb×Yb: 15 mm×5 mm, which is widely used.

[0094] It is desirable that the thermal fuse **60** have a certain size or larger in order to stably function at a low cost. Specifically, it is desirable that the width indicated by Xb in FIG. **10A** be 15 mm or more. Moreover, as a feature of the thermal fuse **60**, the predetermined operating temperature cannot be set to a high temperature. Therefore, Embodiment 2 shows a configuration suitable for a fixing device provided in a low-speed image forming apparatus that does not require operation at relatively high temperatures.

[0095] FIG. **9B** is a schematic diagram showing the installation positions of the thermal fuses **60** of Embodiment 2. As shown in FIG. **9B**, a thermal fuse **60-1** is installed in the heating block HB**21**, a thermal fuse **60-3** is installed in the heating block HB**23**, a thermal fuse **60-4** is installed in the heating block HB**24**, and a thermal fuse **60-6** is installed in the heating block HB**26**. By installing in this way, one or more thermal fuses **60** can be installed for each heating group without being restricted by the device layout.

[0096] Also in Embodiment 2, the combination of the heating blocks HB in which the thermal fuses **60** are installed is not limited to the combination described above. That is, the above-described arrangement is not limiting as long as the thermal fuses **60** are installed between a plurality of heating groups including a plurality of heating blocks HB so as not to be adjacent to each other between the adjacent heating blocks HB. For example, the heating blocks HB in which the thermal fuses **60** are installed may be the heating blocks HB**22**, HB**24**, HB**25**, and HB**27**.

[0097] The thermoswitch **50** shown in Embodiment 1 can also be used as a safety element for the heater **1200** of Embodiment 2. Furthermore, the thermal fuse **60** and the thermoswitch **50** can be used together. Also, in Embodiment 2, as in Embodiment 1, it is possible to insert a resin material or the like between the heater **1200** and the thermal fuse **60**.

[0098] While the present invention has been described with reference to exemplary embodiments, it is to be understood that the invention is not limited to the disclosed exemplary embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.

[0099] This application claims the benefit of Japanese Patent Application No. 2022-025647, filed on Feb. 22, 2022, which is hereby incorporated by reference herein in its entirety.

Claims

1. An image heating device comprising: a heater having a plurality of heating elements arranged in a width direction of a recording material perpendicular to a conveyance direction of the recording material; a nip forming portion that forms a nip where the recording material is nipped; and a control portion that controls power supplied to the plurality of heating elements; wherein the image heating device heats an image formed on the recording material, which is nipped in the nip, by heat of the heater; wherein the plurality of heating elements include a first heating element group and a second heating element group, the first heating element group being arranged symmetrically with respect to a conveyance reference position of the recording material in the width direction and the second heating element group being arranged symmetrically with respect to the conveyance reference position so as to be adjacent to the first heating element group in the width direction; wherein the control portion supplies power to the first heating element group through a first common circuit, and supplies power to the second heating element group through a second common circuit; wherein in the image heating device, the first common circuit includes a first safety element arranged at the heater to respond to the temperature of the heating elements included in the first heating element group; and the second common circuit includes a second safety element arranged at the heater to respond to the temperature of the heating elements included in the second heating element group; wherein the first safety element is arranged on one side with respect to the conveyance reference position in the width direction, and wherein the second safety element is arranged on the other side with respect to the conveyance reference position in the width direction.

2.-8. (canceled)
